



CELL — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this chapter:

1. Cell — Basic Structural & Functional Unit
2. DNA-containing Organelles
3. Viroids, Prions, Mycoplasma & Rickettsia
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6. Plasma Membrane — Composition
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8. Apoptosis — Programmed Cell Death
9. Krebs Cycle & ATP

1. Cell — Basic Structural & Functional Unit

- Cell (कोशिका) life ki structural aur functional unit hai. Simple bacteria se complex animals tak living organisms cells se bane hote hain; cells structure aur respiration, nutrient uptake aur waste removal jaise life processes carry out karte hain. [Q1]

2. DNA-containing Organelles

- Nucleus ke alawa source ke mutabik DNA mitochondria aur chloroplast mein bhi milta hai. MCQ Q2 mein given options ke among correct organelle mitochondrion hai. [Q2]

3. Viroids, Prions, Mycoplasma & Rickettsia

- Viroids short circular single-stranded RNA se bane infectious pathogens hain aur protein coat nahi hota; source mostly plant pathogens batata hai. [Q3]
- Prions proteinaceous infectious particles hain, cells nahi. Mycoplasma cell wall lack karta hai, jabki source Rickettsia mein cell wall present batata hai. [Q3]

4. Plant Cell vs Animal Cell

- Plant cells mein cellulose cell wall hoti hai, animal cells mein cell wall absent hoti hai. Dono plant aur animal cells mein plasma membrane hoti hai. [Q4, Q5, Q6]
- Mature plant cell mein generally ek large central vacuole hoti hai, jabki animal cell mein several small vacuoles ho sakti hain. [Q4]
- Green plant cells mein chloroplasts/green plastids bhi milte hain, jo animal cells mein nahi hote. Plant–animal cell ka main source distinction cell wall ke basis par diya gaya hai. [Q5]

EXAM TRAP — Plant vs Animal Cell

- ▶ Plant cell mein cell wall + plasma membrane dono hote hain; “plant cell has no plasma membrane” statement source mein incorrect hai. [Q4]
- ▶ Animal cell mein cellulose cell wall nahi hoti. [Q5, Q6]

5. Mitochondria — Powerhouse of Cell

- Mitochondrion membrane-bound organelle hai aur respiration ke through ATP generate karta hai; isi liye “powerhouse of the cell” kaha jata hai. [Q7, Q8, Q9]

6. Plasma Membrane — Composition

- Plasma membrane/cell membrane cell ke interior ko outside environment se separate karti hai. Source ke mutabik ye mainly lipid (phospholipid) bilayer aur embedded proteins se bani hoti hai. [Q10]

7. Protein Synthesis — Rough Endoplasmic Reticulum

- Rough endoplasmic reticulum (RER) ki outer membrane par ribosomes present hote hain; source Q11 mein RER ko protein synthesis mein most significant role wala organelle answer karta hai. [Q11]

8. Apoptosis — Programmed Cell Death

- Apoptosis programmed cell death (PCD) ka process hai. Source ise multicellular organisms aur kuch eukaryotic single-celled microorganisms (jaise yeast) mein describe karta hai; old, unnecessary aur unhealthy cells eliminate karne mein role batata hai. [Q12]

9. Krebs Cycle & ATP

- Krebs cycle enzyme-catalyzed reactions ka cycle hai jo aerobic metabolism se related hai; eukaryotic cell mein mitochondrial matrix mein occur karta hai aur source Hans Krebs (1937) se associate karta hai. [Q13]
- Source Q13 ke answer/explanation ke mutabik Krebs cycle system mein maximum number of ATP molecules generate karne wala term diya gaya hai. [Q13]

SOURCE-SPECIFIC NOTE — Q13

- Q13 ka ATP-related statement yahan source HTML ki wording/explanation ke according preserve kiya gaya hai; is mind map mein outside correction/verification add nahi ki gayi. [Q13]